A Refined Maximum-Likelihood Inspired Tomographic SAR Imaging in High Noise Level

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***Abstract*—We consider using the maximum-likelihood (ML)-inspired methods, such as ML-inspired adaptive robust iterative approach (MARIA), for tomographic synthetic aperture radar (TomoSAR) imaging in scenarios with high noise level. Traditional MARIA method faces two key challenges in this case: first, the positive feedback mechanism likely leads to error propagation in high noise level; second, the assumption of exactly known noise power is usually invalid, both of which result in significant performance degradation. Therefore, we propose a refined ML-inspired TomoSAR imaging method suitable for high noise level, improving the signal and noise estimation process of the typical MARIA. First, we derive a new iterative expression based on ML criterion without the positive feedback mechanism for signal estimation, which suppresses the error propagation in high noise level. Second, we regard noise power as a variable to be estimated and introduce an iterative method based on ML criterion for estimation. The simulation and real unmanned aerial vehicle (UAV) experiment results both verify the effectiveness of the proposed method. Finally, we theoretically give a convergence guarantee for the proposed method.**

***Index Terms*—Convergence guarantee, high noise level, maximum likelihood (ML), tomographic synthetic aperture radar (TomoSAR).**

I. Introduction

TOMOGRAPHIC synthetic aperture radar (TomoSAR) [1], [2], [3] is a technique for achieving 3-D synthetic aperture radar (SAR) imaging. Traditional 2-D SAR imaging suffers from the issue of layover [4] that all targets within the same range cell are superimposed onto a single pixel, which results in ambiguity among targets at different heights. By constructing a synthetic aperture in elevation through multiple tracks, TomoSAR is capable of separating targets at different heights and effectively addresses the layover issue. This technique shows great potential in applications, such as terrain mapping [5], [6], biomass estimation [7], [8], forest cover analysis [9], [10], [11], 3-D imaging of building structures [12], [13], [14], and other civilian and military uses.

The core task of TomoSAR, compared to traditional 2-D SAR imaging, is the elevation estimation of the target scene, which is essentially a harmonic estimation problem. Its specificity lies in the fact that the scene is usually observed with low signal-to-noise ratio (SNR). The low SNR is due to factors, such as the weak scattering from foliage in forested areas, which reduces the SNR of the received echoes [15]. Additionally, both the radar weight and volume constraints and long-range detection contribute to a reduction in the SNR [16].

Several methods are proposed for solving the harmonic recovery problem, including matched filtering (MF) methods [17], subspace-based methods [18], compressed sensing (CS) methods [19], [20], [21], maximum-likelihood (ML) methods [22], [23], [24], [25], and so on. The MF method is a technique used to extract signals from noise by maximizing the SNR. It works by correlating the received signal with a reference signal that matches the expected shape of the signal of interest. This method is simple and does not require additional prior assumptions. However, it is sensitive to the SNR and usually has poor resolution with high side lobes. The subspace method is a technique by exploiting the underlying structure of the signal space. It involves decomposing the signal into signal and noise subspaces to separate and estimate the signal components. This method is simple to implement and can achieve super-resolution, but it requires an accurate prior assumption of the division of signal and noise subspace. Also, it relies on a high SNR to construct an accurate signal subspace. The CS method is a technique used to recover signals by exploiting the sparsity, which means that the signal can be represented by a few nonzero coefficients in a suitable basis or dictionary. This method achieves super-resolution under low SNR, but the iterative solving process is usually slow to converge, and its performance heavily depends on the sparsity prior assumption. The ML method is a technique that estimates parameters by maximizing the likelihood function, representing the probability of the observed data given the parameters. It typically requires to impose assumptions on the probability distribution of the estimated variables and achieves optimal performance under the assumptions. This method can theoretically achieve super-resolution under low SNR for both sparse and nonsparse scenarios, but it involves complex computations, and its optimal performance depends on relatively accurate prior assumptions. Given the specificity of the TomoSAR scenarios, this article focuses on using the ML methods, which have the potential to achieve super-resolution in nonsparse scenarios. Among them, ML-inspired adaptive robust iterative approach (MARIA) [25] and iterative adaptive approach (IAA) (including robust IAA) [26], [27] are

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typical examples of the ML technique in TomoSAR imaging. Particularly, according to the discussions in [24], IAA can be regarded as a variant of MARIA, where MARIA modifies the iteration, enhancing the resolution. The MARIA algorithm iteratively estimates the signal power and noise power, seeking the target parameters that maximize the likelihood function. This method exhibits super-resolution performance and has been successfully applied in both urban and forest scenarios [28], [29].

However, the typical MARIA algorithm faces two major challenges under low SNR scenarios, which significantly affect parameter estimation performance.

1) *Error Propagation:* The typical MARIA has a positive feedback mechanism, which acts as a prior weight for subsequent iterations. This means that when the initial iteration values are relatively accurate, the algorithm can accelerate convergence to the true values by filtering out the erroneous estimates during the iteration. However, under low SNR conditions, the initial iteration values are hardly accurate, which can lead to the true values being misidentified and filtered out. Once this error occurs, it cannot be corrected in subsequent iterations, resulting in error propagation.
2) *Mismatched Noise Model:* The typical MARIA assumes that the noise power for each acquisition is either known or accurately pre-estimated. However, in high noise level scenarios, the noise powers across multiple acquisitions are rarely known with large variance, leading to a mismatch in the error model.

Both of the two factors impact the performance of MARIA. Additionally, the typical MARIA employs an iterative solving approach but lacks a guarantee of convergence, which makes it challenging to effectively guide practical application.

By improving the existing MARIA algorithm, we propose a new TomoSAR imaging method based on the ML technique, which achieves better performance under low SNR conditions. Particularly, we first rederive the iteration expression without the positive feedback mechanism, which avoids the error propagation issue in high noise level. Then, we perform an iterative noise power estimation based on ML criterion rather than relying on a prior assumption, which avoids noise model mismatch under high noise level conditions. The feasibility of the proposed method is verified by both simulation and real unmanned aerial vehicle (UAV) experimental results. Finally, we provide a theoretical convergence guarantee for the proposed method based on the fixed-point theorem and analyze the conditions for its application.

The main contributions of this article are listed as follows.

1) *Algorithm Refinement:* Based on the MARIA method, we improve the signal power and noise power estimation styles by rederiving the signal iteration expression without the positive feedback mechanism and supplementing the iterative noise power estimation, respectively, enabling better SAR elevation estimation results under low SNR scenarios. The effectiveness of improvement is verified by simulation and real UAV experimental results. This greatly expands the applicability of the ML methods in practical scenarios.
2) *Convergence Analysis:* We provide a theoretical guarantee for the convergence of the proposed ML method, whereas the typical MARIA lacks such a convergence analysis. Particularly, based on the assumptions of column orthogonality of the observation matrix and identical noise power across different data acquisitions, we use the fixed-point theorem for derivation and proof. The rationality of these assumptions is also explained.

The rest of this article is organized as follows. Section II introduces the signal model of TomoSAR and challenges of MARIA. The proposed algorithm for high noise level is detailed in Section III. A theoretical analysis of the proposed method on convergence is given in Section IV. We show the simulation and real UAV experiment results to verify the advantages of the proposed method in Section V. Section VI concludes this article.

Symbol Denotation: We use $\mathbb{C}$ to denote the set of complex numbers. Uppercase boldface letters denote matrices (e.g., $\mathbf{A}$) and lowercase boldface letters denote vectors (e.g., $\mathbf{a}$). The (m, n)th element of a matrix $\mathbf{A}$ is denoted by $[\mathbf{A}]_{m,n}$, and the nth column is represented by $[\mathbf{A}]_n$. The superscript $(\cdot)^+$ denotes the Hermitian conjugate operation on matrices, $D(\cdot)$ represents the diagonalization operator that transforms a vector into a diagonal matrix, and $\{\cdot\}_{\text{diag}}$ refers to the operator that extracts the diagonal elements of a matrix into a vector. We denote the determinant of a matrix by $\det(\cdot)$ and the trace of a matrix by $\text{trace}(\cdot)$. A statistical averaging operator is denoted by $\langle\cdot\rangle$. The amplitude of a scalar and the l_2 norm of a vector are represented by $|\cdot|$ and $\|\cdot\|_2$, respectively. We denote the probability by $p(\cdot)$ and the imaginary unit for complex numbers by $j = \sqrt{-1}$.

II. Challenges of MARIA

In this section, we first introduce ML-based TomoSAR imaging signal model in Section II-A. Then, we discuss the challenges associated with the MARIA method and highlight the motivation behind optimizing the algorithm in Section II-B.

A. Signal Model

Consider L SAR images acquired by L tracks with different heights. Each track is called a baseline in TomoSAR, and the baselines are denoted by $\{b_l\} = \{b_1, b_2, \ldots, b_L\}$. We regard the center baseline as the master baseline, whose position is set as zero. The overall baseline span along perpendicular to line-of-sight direction, also known as elevation, is denoted by Δb. The geometry of TomoSAR imaging is shown in Fig. 1.

Consider the lth baseline toward the target scene. The observation signal is an integral of all scatterers located along an arc with the same slant range r_l. When the targets are far from the radar, the equidistance arc can be approximated as a straight-line axis along elevation direction, denoted by s. The observation signal of the lth baseline is expressed as

$$g_l = \int \gamma(s) \exp\left(-j\frac{4\pi r_l}{\lambda}\right) ds + \nu_l \tag{1}$$

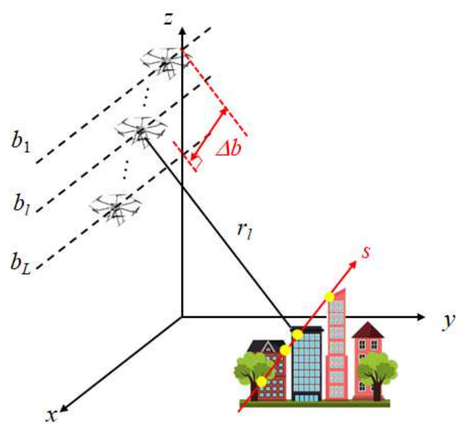

Fig. 1. Geometry of TomoSAR.

where λ denotes the wavelength corresponding to the signal center frequency f_c, $\gamma(s)$ denotes the scattering coefficient in elevation, and ν_l is the noise for $l = 1, \ldots, L$.

After deramping in elevation, (1) can be written as

$$g_l = \int \gamma(s) \exp\left(-j\frac{4\pi}{\lambda}\frac{b_l s}{r}\right) ds + \nu_l \tag{2}$$

where $l = 1, \ldots, L$, and r denotes the slant range from master baseline to the targets. Then, (2) can be transformed into matrix form

$$\mathbf{g} = \mathbf{A}\mathbf{s} + \mathbf{n} \tag{3}$$

provided that the elevation axis s is discretized into M grids, where $\mathbf{g} = [g_1, \ldots, g_L]^{\mathrm{T}} \in \mathbb{C}^{L\times 1}$ is the data measurement, $\mathbf{s} = [\gamma(s_1), \ldots, \gamma(s_M)]^{\mathrm{T}} \in \mathbb{C}^{M\times 1}$ is the scattering coefficient vector, $\mathbf{n} = [\nu_1, \ldots, \nu_L]^{\mathrm{T}} \in \mathbb{C}^{L\times 1}$ is the noise vector, and $\mathbf{A} \in \mathbb{C}^{L\times M}$ is the measurement matrix, whose (l, m)th element is given by

$$[\mathbf{A}]_{l,m} = \exp\left(-j\frac{4\pi b_l s_m}{\lambda r}\right). \tag{4}$$

The elevation resolution, similar to the azimuth resolution in SAR [30], is defined as

$$\rho = \frac{\lambda r}{2\Delta b}. \tag{5}$$

Especially, when the baselines are uniformly spaced, the unambiguous imaging range is

$$H_{\mathrm{amb}} = L\rho. \tag{6}$$

In practical scenarios, the number of targets N_t is smaller than the number of grids M, implying that the vector $\mathbf{s}$ contains only a small fraction of nonzero elements. In this case, N_t may be sufficiently small to render the scenario sparse, or it may be comparable to the number of baselines L, resulting in a nonsparse scenario. Under such conditions, ML-based method exhibits the optimum performance. Compared to the MF method, the ML method achieves super-resolution, and compared to subspace method, it is less sensitive to noise. Additionally, it does not require the sparsity assumption present in CS. In the scenes where the target structures are not necessarily sparse (e.g., forests), ML-inspired method can outperform other approaches.

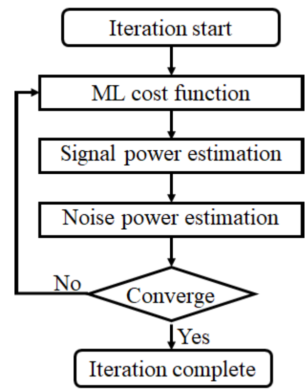

Fig. 2. Flowchart of MLOS.

To solve (3), MARIA, a representative ML-based method is proposed. This ML method assumes that $\mathbf{s}$ and $\mathbf{n}$ are independent zero-mean complex Gaussian variables, yielding that $\mathbf{g}$ is also a zero-mean complex Gaussian variable. For real SAR systems, noise originates from various sources, including both independent Gaussian noise and correlated noise, such as crosstalk and platform vibrations. Correlated noise can typically be suppressed or compensated. Crosstalk is usually suppressed beforehand during system implementation, while platform vibrations can be mitigated through motion compensation or autofocus techniques to avoid image defocusing. As a result, in low SNR scenarios where SNR is limited below 20 dB, the impact of correlated noise is generally subordinate to that of uncorrelated noise. Therefore, we model the noise vector $\mathbf{n}$ as a vector with independent components. The cost function of MARIA is the likelihood probability of $\mathbf{g}$, given by

$$p(\mathbf{g}) = (2\pi)^{-\frac{L}{2}} \det{}^{-\frac{1}{2}}\left(\mathbf{R}_{\mathbf{g}}\right)\ \exp\left(-\frac{1}{2}\mathbf{g}^{+}\mathbf{R}_{\mathbf{g}}^{-1}\mathbf{g}\right) \tag{7}$$

where $\mathbf{R}_{\mathbf{g}}$ is the covariance matrix of $\mathbf{g}$, expressed as

$$\mathbf{R}_{\mathbf{g}} = \langle \mathbf{g}\mathbf{g}^{+} \rangle = \mathbf{A}\mathbf{R}_{\mathbf{s}}\mathbf{A}^{+} + \mathbf{R}_{\mathbf{n}}. \tag{8}$$

The goal of MARIA is to recover $\mathbf{s}$ by iteratively solving the maximization of (7), which is achieved through the joint estimation of signal covariance matrix $\mathbf{R}_{\mathbf{s}}$ and noise covariance matrix $\mathbf{R}_{\mathbf{n}}$. Under the assumption of independence on $\mathbf{s}$ and $\mathbf{n}$, both $\mathbf{R}_{\mathbf{s}}$ and $\mathbf{R}_{\mathbf{n}}$ are diagonal matrices. Therefore, the joint estimation of $\mathbf{R}_{\mathbf{s}}$ and $\mathbf{R}_{\mathbf{n}}$ is equivalent to that of the power of $\mathbf{s}$ and $\mathbf{n}$, yielding the desired estimation of $\mathbf{s}$.

B. Challenge and Motivation

However, the typical MARIA algorithm has significant shortcomings in practical applications, manifested as two aspects: 1) a noticeable performance degradation in the scenarios with high noise level, such as the forest scenario and far detection range cases and 2) the iterative solving algorithm of the typical MARIA lacks theoretical guarantees for convergence. We detail these challenges as follows.

For the first aspect, typical MARIA algorithm faces error propagation issues and mismatched noise model problems under high noise level conditions, resulting in a significant performance degradation.

For error propagation, it is noted that the iterative solving steps of typical MARIA algorithm include a weighting matrix, which, through positive feedback, accelerates the iteration speed and enhances super-resolution capability when the algorithm's initial value or intermediate estimates are close to the true values. However, under high noise level, the initial value or intermediate estimates of the algorithm often deviate significantly from the true values, and the weighting matrix can have a negative effect by driving the results further away from the true values, called error propagation. In particular, when the power estimate of an element of $\mathbf{s}$ is zero during the iteration, it will remain zero in subsequent iterations. This results in the failure to accurately recover that target component if, during any iteration, the true counterpart is mistakenly set to zero. Consequently, the error propagation will significantly reduce the accuracy and robustness of the estimation results of typical MARIA under high noise level.

For mismatched noise model, the typical MARIA assumes that the noise covariance is known or can be accurately estimated. However, in the high noise level case, the noise covariance is difficult to obtain or estimate precisely, leading to a mismatch in the noise model and affecting the algorithm's performance.

For the second aspect, typical MARIA algorithm is based on iterative solving methods, which involve complex matrix inversion steps, making it difficult to intuitively assess the algorithm's performance. Existing ML-inspired methods lack theoretical guarantees of convergence and require further in-depth research and analysis to determine the performance boundaries of the algorithms.

To meet the above challenges, our main goals are: 1) to propose a TomoSAR imaging algorithm for the scenarios with high noise level, which is based on an ML-inspired framework and 2) to provide a theoretical guarantee on the convergence of the proposed algorithm.

III. Algorithm

In this section, we introduce a new method for TomoSAR imaging, referred to as MARIA for low SNR (MLOS). The proposed MLOS method consists of two main components: one is the estimation of signal power, and the other is the estimation of noise power. First, we present the main algorithm framework and the cost function in Section III-A. Then, we introduce the signal power estimation and noise power estimation of MLOS in Sections III-B and III-C, respectively, where each is followed by a comparison with typical MARIA.

A. Main Framework and Cost Function

The main framework of the proposed MLOS method adopts an alternating iterative approach, performing sequential estimation of signal power and noise power. Particularly, iterations are executed in two steps. The first step is the signal power estimation and the second step is the noise power estimation, which are both based on the ML criterion. The iteration terminates when the estimate of signal power in the algorithm converges. The flowchart of MLOS is presented in Fig. 2.

Based on the assumptions that $\mathbf{s}$ and $\mathbf{n}$ are independent zero-mean Gaussian variables, we have

$$\mathbf{R_s} = \left\langle \mathbf{s}\mathbf{s}^{+} \right\rangle = D(\mathbf{v}) \tag{9}$$

$$\mathbf{R_n} = \left\langle \mathbf{n}\mathbf{n}^{+} \right\rangle = D(\mathbf{u}) \tag{10}$$

which means that $\mathbf{R_s}, \mathbf{R_n}$ are the diagonal matrices, and $\mathbf{v}, \mathbf{n}$ represent the signal and noise power, respectively. Then, the covariance of $\mathbf{g}$ in (8) can be recast as

$$\mathbf{R_g}(\mathbf{v},\mathbf{u}) = \mathbf{A}D(\mathbf{v})\mathbf{A}^{+} + D(\mathbf{u}). \tag{11}$$

Thus, the log-likelihood cost function of (7) is recast as

$$\begin{aligned}\mathcal{L}(\mathbf{v},\mathbf{u}) &= \ln\left(p(\mathbf{g};\mathbf{v},\mathbf{u})\right)\\ &= -\frac{1}{2}\left(\ln\left(\det\left(\mathbf{R_g}\right)\right) + \mathbf{g}^{+}\mathbf{R_g}^{-1}\mathbf{g} + L\ln 2\pi\right).\end{aligned} \tag{12}$$

B. Estimation of Signal Power v

The signal power $\mathbf{v}$ is estimated by solving the maximization of the ML cost function $\mathcal{L}(\mathbf{v},\mathbf{u})$ in (12) with respect to $\mathbf{v}$ when $\mathbf{u}$ is fixed, given by

$$\widehat{\mathbf{v}} = \underset{\mathbf{v}}{\operatorname{argmax}}\ \mathcal{L}(\mathbf{v},\mathbf{u}). \tag{13}$$

Which is to find the zero point of the first-order derivative of (13)

$$\frac{\mathrm{d}\mathcal{L}(\mathbf{v},\mathbf{u})}{\mathrm{d}\mathbf{v}} = 0. \tag{14}$$

Based on the following derivative expressions:

$$\frac{\mathrm{d}\ln\left(\det\left(\mathbf{R_g}\right)\right)}{\mathrm{d}\mathbf{v}} = \left\{\mathbf{A}^{+}\mathbf{R_g}^{-1}\mathbf{A}\right\}_{\text{diag}} \tag{15}$$

$$\frac{\mathrm{d}\ \mathbf{g}^{+}\mathbf{R_g}^{-1}\mathbf{g}}{\mathrm{d}\mathbf{v}} = -\left\{\mathbf{A}^{+}\mathbf{R_g}^{-1}\mathbf{g}\mathbf{g}^{+}\mathbf{R_g}^{-1}\mathbf{A}\right\}_{\text{diag}} \tag{16}$$

we recast (14) as

$$\left\{\mathbf{A}^{+}\mathbf{R_g}^{-1}\mathbf{g}\mathbf{g}^{+}\mathbf{R_g}^{-1}\mathbf{A}\right\}_{\text{diag}} = \left\{\mathbf{A}^{+}\mathbf{R_g}^{-1}\mathbf{A}\right\}_{\text{diag}}. \tag{17}$$

To ensure consistency between the forms on both sides of (17), we rewrite it as

$$\left\{\mathbf{A}^{+}\mathbf{R_g}^{-1}\mathbf{g}\mathbf{g}^{+}\mathbf{R_g}^{-1}\mathbf{A}\right\}_{\text{diag}} = \left\{\mathbf{A}^{+}\mathbf{R_g}^{-1}\mathbf{R_g}\mathbf{R_g}^{-1}\mathbf{A}\right\}_{\text{diag}}. \tag{18}$$

Denote a weighting matrix $\mathbf{F} \in \mathbb{C}^{M\times L}$ by

$$\mathbf{F} = \mathbf{A}^{+}\mathbf{R_g}^{-1} \tag{19}$$

and simplify (18) as

$$\left\{\mathbf{F}\mathbf{g}\mathbf{g}^{+}\mathbf{F}^{+}\right\}_{\text{diag}} = \left\{\mathbf{F}\mathbf{R_g}\mathbf{F}^{+}\right\}_{\text{diag}}. \tag{20}$$

By substituting (11) to the right part of (20), we have

$$\left\{\mathbf{F}\mathbf{A}D(\mathbf{v})\mathbf{A}^{+}\mathbf{F}^{+}\right\}_{\text{diag}} = \left\{\mathbf{F}\left(\mathbf{g}\mathbf{g}^{+} - D(\mathbf{u})\right)\mathbf{F}^{+}\right\}_{\text{diag}}. \tag{21}$$

Since $D(\mathbf{v})$ is a diagonal matrix, we have

$$\left\{\mathbf{F}\mathbf{A}D(\mathbf{v})\mathbf{A}^{+}\mathbf{F}^{+}\right\}_{\text{diag}} = \mathbf{P}\mathbf{v} \tag{22}$$

where $\mathbf{P} = D(\{\mathbf{F}\mathbf{A}\mathbf{A}^{+}\mathbf{F}^{+}\}_{\text{diag}})$. Substituting (22) to (21) implies the estimation of $\mathbf{v}$, given by

$$\mathbf{v} = \mathbf{P}^{-1}\left\{\mathbf{F}\left(\mathbf{g}\mathbf{g}^{+} - D(\mathbf{u})\right)\mathbf{F}^{+}\right\}_{\text{diag}}. \tag{23}$$

Note that $\mathbf{F}$ and $\mathbf{P}$ are solution-dependent, as both include the component $\mathbf{R_g^{-1}}$, which in turn contains $\mathbf{v}$ and $\mathbf{u}$. This implies that obtaining a closed-form solution for $\mathbf{v}$ is difficult, and an iterative method is required. At the beginning of the iteration, $\mathbf{v}$ and $\mathbf{u}$ are first initialized as $\mathbf{v}^{[0]}$ and $\mathbf{u}^{[0]}$, respectively. At the kth iteration, the signal power $\mathbf{v}^{[k]}$ is updated by

$$\mathbf{v}^{[k]} = \mathbf{P}^{[k]-1}\left\{\mathbf{F}^{[k]}\left(\mathbf{G} - D\left(\mathbf{u}^{[k-1]}\right)\right)\mathbf{F}^{[k]+}\right\}_{\text{diag}} \tag{24}$$

where $\mathbf{F}^{[k]} = \mathbf{A}^{+}\mathbf{R}_{\mathbf{g}}^{-1}(\mathbf{v}^{[k-1]}, \mathbf{u}^{[k-1]})$, and $\mathbf{G}$ is the estimated covariance matrix of $\mathbf{g}$, given by

$$\mathbf{G} = \frac{1}{I}\sum_{i=1}^{I}\mathbf{g}(i)\,\mathbf{g}(i)^{+} \tag{25}$$

and I is the number of independent realizations, which represents the multiple snapshots or looks.

Remark 1: The main difference between the proposed MLOS and typical MARIA lies in the expression of weighting matrix $\mathbf{F}$. In typical MARIA, the weighting matrix is $\mathbf{F} = D(\mathbf{v})\mathbf{A}^{+}\mathbf{R}_{\mathbf{g}}^{-1}$, which has an additional positive feedback weight matrix with respect to $\mathbf{v}$, $D(\mathbf{v})$, compared to (19) in MLOS, and the corresponding derivations are also different. The positive feedback matrix $D(\mathbf{v})$ causes error propagation, especially when the noise level is high, such that the initial values and intermediate estimates of $\mathbf{v}$ differ greatly from the true values. In an extreme case, if an intermediate estimate satisfies $[\mathbf{v}^{[k-1]}]_m = 0$, we have $[\mathbf{F}^{[k]}]_m = 0$ in typical MARIA, thus $[\mathbf{v}^{[k]}]_m = 0$ in (24), leading to no update for that component of $\mathbf{v}$. This means that if the true value is mistakenly set to zero during any iteration, it will never be correctly estimated in subsequent iterations, causing the error to accumulate and propagate without correction. In MLOS, we perform the derivation approach without the positive feedback weight matrix with respect to $\mathbf{v}$, thereby avoiding the issue of error propagation. The effectiveness of the refinement is validated by simulations.

C. Estimation of Noise Power $\mathbf{u}$

In MLOS, we consider estimating the noise power $\mathbf{u}$ by solving the maximization of the ML cost function $\mathcal{L}(\mathbf{v}, \mathbf{u})$ in (12) with respect to $\mathbf{u}$ when $\mathbf{v}$ is fixed, given by

$$\widehat{\mathbf{u}} = \underset{\mathbf{u}}{\operatorname{argmax}}\ \mathcal{L}(\mathbf{v}, \mathbf{u}) \tag{26}$$

which is to find the zero point of the derivative of (26)

$$\frac{\mathrm{d}\mathcal{L}(\mathbf{v}, \mathbf{u})}{\mathrm{d}\mathbf{u}} = 0. \tag{27}$$

By calculation, we have

$$\left\{\mathbf{R}_{\mathbf{g}}^{-1}\mathbf{g}\mathbf{g}^{+}\mathbf{R}_{\mathbf{g}}^{-1}\right\}_{\text{diag}} = \left\{\mathbf{R}_{\mathbf{g}}^{-1}\right\}_{\text{diag}}. \tag{28}$$

Similar with the derivation of signal power estimation in Section III-B, the iterative expression of $\mathbf{u}$ is directly shown as follows:

$$\mathbf{u} = \mathbf{Q}^{-1}\left\{\mathbf{R}_{\mathbf{g}}^{-1}\left(\mathbf{g}\mathbf{g}^{+} - \mathbf{A}D(\mathbf{v})\mathbf{A}^{+}\right)\mathbf{R}_{\mathbf{g}}^{-1}\right\}_{\text{diag}} \tag{29}$$

where $\mathbf{Q} = D(\{\mathbf{R}_{\mathbf{g}}^{-2}\}_{\text{diag}})$. Particularly, at the kth iteration, the noise power $\mathbf{u}^{[k]}$ is updated by

$$\mathbf{u}^{[k]} = \mathbf{Q}^{[k]-1}\left\{\mathbf{R}_{\mathbf{g}}^{[k]-1}\left(\mathbf{G} - \mathbf{A}D\left(\mathbf{v}^{[k]}\right)\mathbf{A}^{+}\right)\mathbf{R}_{\mathbf{g}}^{[k]-1}\right\}_{\text{diag}} \tag{30}$$

where $\mathbf{R}_{\mathbf{g}}^{[k]-1} = \mathbf{R}_{\mathbf{g}}^{-1}(\mathbf{v}^{[k]}, \mathbf{u}^{[k-1]})$.

During the noise iteration process, outliers may occasionally occur, where the estimates deviate significantly from the true values, making it difficult to converge. To address this issue, we apply the constant false alarm rate principle during noise power estimation to eliminate outliers. Considering that the noise follows a Gaussian distribution and its power is nonnegative, we estimate the distribution of the noise power, extract the outliers based on constant false alarm rate, and remove them accordingly. This improves the stability of the noise power estimation process, ensuring the overall convergence. If prior knowledge of the noise distribution is available, some safeguards, such as prior regularization or minimum noise floors, can be further introduced into the estimation process. These methods can be directly incorporated as improvements to the derivation of the proposed method, such as [31] and [32]. This not only ensures the estimation performance of the proposed method in the absence of prior knowledge but also allows the incorporation of prior information to enhance performance.

Remark 2: The difference between the proposed MLOS and typical MARIA lies in that whether the noise power $\mathbf{u}$ is estimated iteratively based on the received signals or preassumed to be known. Particularly, typical MARIA assumes that $\mathbf{u}$ is known or well pre-estimated. This is difficult to achieve in high noise level scenarios because the large variance of noise power, leading to noise model mismatch in typical MARIA. The proposed MLOS effectively addresses this issue through an iterative joint estimation on $\mathbf{v}$ and $\mathbf{u}$.

In summary, the MLOS method iteratively carries out the estimation of signal power $\mathbf{v}$ in (24) and noise power $\mathbf{u}$ in (30). Beforehand, $\mathbf{v}$ and $\mathbf{u}$ are initialized as the conventional beamforming $\mathbf{v} = \{\mathbf{A}^{+}\mathbf{G}\mathbf{A}\}_{\text{diag}}$ and identity matrix $\mathbf{u} = N_0\mathbf{I}$, where N_0 is the noise power characterized by system SNR. The iteration terminates when the difference between the adjacent estimates of $\mathbf{v}$ is less than a predefined threshold $\mathcal{T}$ or the iteration reaches the maximum times $K_{\max}$. In the first case, we use the l_2 norm as the metric and the termination criterion is given by

$$\left\|\mathbf{v}^{[k]} - \mathbf{v}^{[k-1]}\right\|_2 \leqslant \mathcal{T}. \tag{31}$$

Algorithm 1 MLOS

Input: Received signals $\mathbf{g}$, measurement matrix $\mathbf{A}$, initialization $\mathbf{v}^{[0]}, \mathbf{u}^{[0]}$, iteration index $k = 1$, threshold $\mathcal{T}$, and maximum iteration times $K_{\max}$.
  Calculate the covariance matrix $\mathbf{G}$ in (25).
  while $\|\mathbf{v}^{[k]} - \mathbf{v}^{[k-1]}\|_2 > \mathcal{T}$ and $k \leqslant K_{\max}$ **do**
    (1) Update the signal power $\mathbf{v}^{[k]}$ as (24).
    (2) Update the noise power $\mathbf{u}^{[k]}$ as (30).
  end while
Output: The signal power estimation $\widehat{\mathbf{v}}$.

The selection of the threshold $\mathcal{T}$ is typically related to the elevation grid density, with denser grids requiring a larger threshold $\mathcal{T}$. This is because increased grid density tends to degrade the orthogonality of matrix $\mathbf{A}$, thereby amplifying the effect of perturbations in the estimation process and making the convergence condition more difficult to satisfy. Accordingly, in this study, the threshold is formulated as $\mathcal{T} = \epsilon M\rho/\Delta s$, where Δs denotes the elevation grid density, and ϵ, set to 10^{-4}, serves as the convergence tolerance. The maximum iteration $K_{\max}$ is set to be 15, as convergence is generally observed at $k = 10$ according to [25] and [24]; a margin is therefore reserved by choosing 15 in this study. The whole MLOS method is summarized as Algorithm 1.

The overall algorithmic complexity can be expressed as the product of the total number of iterations k and the computational cost per iteration. Each iteration of the proposed method involves two main components: the estimation of signal power $\mathbf{v}$ and noise power $\mathbf{u}$. For the computation of $\mathbf{v}$, the primary step is obtaining the inverse of the covariance matrix ${\mathbf{R_g}}^{-1}$, which requires $\mathcal{O}(L^3)$ floating point operations (FLOPs). The subsequent computation of matrix $\mathbf{F}$ requires $\mathcal{O}(ML^2)$ FLOPs, while the matrix $\mathbf{P}$ demands $\mathcal{O}(M^3)$ FLOPs. Furthermore, evaluating the term $\{\mathbf{F}(\mathbf{G}-\mathbf{D}(\mathbf{u}))\mathbf{F}^+\}$ requires $\mathcal{O}(ML^2+M^2L)$ FLOPs. The total computational cost of signal power estimation $\mathbf{v}$ is $\mathcal{O}(L^3 + ML^2 + M^2L + M^3)$ FLOPs. For noise power estimation $\mathbf{u}$, the previously computed ${\mathbf{R_g}}^{-1}$ can be reused.The computation of $\mathbf{Q}$ requires $\mathcal{O}(L^3)$ FLOPs and the term in the curly brace of (30) requires $\mathcal{O}(L^3 + ML^2)$ FLOPs. The total computational cost of noise power estimation $\mathbf{u}$ is $\mathcal{O}(L^3 + ML^2)$ FLOPs. In summary, the per-iteration complexity of the MLOS is $\mathcal{O}(L^3 + ML^2 + M^2L + M^3)$. Compared to MARIA, MLOS removes the positive feedback mechanism, saving a negligible $\mathcal{O}(M^2)$ FLOPs, and introduces an extra $\mathcal{O}(L^3 + ML^2)$ FLOPs, without adding computational burden in terms of higher order complexity.

IV. Theoretical Analysis on Convergence

In this section, we present a theoretical analysis on the convergence of the proposed MLOS method. Particularly, under the assumption that the columns of the observation matrix are orthogonal and the noise power is the same for different tracks, we provide a theoretical guarantee on the convergence of MLOS in Section IV-A. Then, we analyze the property of approximate column orthogonality of the observation matrix in practical TomoSAR scenarios in Section IV-B.

A. Theoretical Guarantee on Convergence

In this subsection, we present a theoretical guarantee on the convergence of MLOS method under certain assumptions, given by

Proposition 1: When $\mathbf{A} \in \mathbb{C}^{L\times M}$ is a column orthogonal matrix, such that $\mathbf{A}^+\mathbf{A} = L\mathbf{I}$ with $L > M$ and the covariance matrix of $\mathbf{u}$ is identical, i.e., $D(\mathbf{u}) = \sigma^2\mathbf{I}$, the iteration of $\mathbf{v}$ in MLOS method converges to a unique solution.

Proof: See Appendix A.■

In the TomoSAR scenarios, Proposition 1 indicates that when the measurement matrix is column orthogonal and the noise power of each observation is identical, MLOS method to recover the elevation information can converge. The orthogonal assumption is discussed in Section IV-B, and the noise assumption is used for the convenience of the proof, which will be further relaxed in the future work.

Note that, in practice, the measurement matrix $\mathbf{A}$ is row full rank with $M > L$, and the estimation can be viewed as a process of basis selection of $\mathbf{A}$ corresponding to the real target signals. However, the number of true basis in $\mathbf{A}$, N_t, usually satisfies $N_t < L$. For brevity, we only consider the estimation on the true signal and let $M = N_t < L$ in the proposition.

B. Approximate Column Orthogonality

In this subsection, we show the approximate orthogonality between the columns of the measurement matrix $\mathbf{A}$, given by

Proposition 2: When $b_l \sim \mathcal{U}[0, D], n = 1, \ldots, L$, for any $t > 0$, define t_1, t_2 as

$$t_{m_1} \equiv \max\left\{\left|t + \frac{\sin\Delta\omega D}{\Delta\omega D}\right|, \left|-t + \frac{\sin\Delta\omega D}{\Delta\omega D}\right|\right\} \tag{32}$$

$$t_{m_2} \equiv \left|t + \frac{1-\cos\Delta\omega D}{\Delta\omega D}\right| \tag{33}$$

where $\Delta\omega = 4\pi\Delta s/\lambda r$ and $\Delta s = s_{m_1} - s_{m_2}$ for $m_1, m_2 = 1, \ldots, M$, $m_1 \neq m_2$. When $\Delta\omega \geqslant 2\pi/D$, we have

$$p\left(|\frac{1}{L}\left[\mathbf{A}^+\mathbf{A}\right]_{m_1,m_2}| \geqslant \sqrt{t_1^2 + t_2^2}\right) \leqslant 4e^{-\frac{Lt^2}{2}}. \tag{34}$$

Proof: See Appendix B.■

Specifically, when two targets are separated by one resolution in elevation, i.e., $\Delta s = \rho$ in (5), we have $\Delta\omega D = 2\pi$. In this case, the probability that $(1/L)[\mathbf{A}^+\mathbf{A}]_{m_1,m_2} \geqslant 0.25$ is less than 0.25 based on Proposition 2 when the number of baselines is $L = 177$, which indicates that the two columns are likely uncorrelated, yielding approximate orthogonality. The high requirement on the number of L is due to the use of scaling in the proof to avoid extreme cases. In practice, such a large L is typically not necessary.

Also, it should be noted that the above derivations are based on the assumptions of column orthogonality of matrix $\mathbf{A}$ and uniform noise power. In practice, these assumptions are not strictly satisfied, which may lead to incomplete convergence of the iterative process. Overall, empirical convergence is observed even when assumptions are relaxed, which is analyzed in detail in Section V.

V. Simulations and Experiments

In this section, we use the simulations and real UAV experiments to verify the feasibility of the proposed MLOS method. Section V-A introduces the simulation settings. Section V-B shows the comparison between MLOS, typical MARIA, and CS methods. Section V-C presents the results of the real UAV TomoSAR experiment.

A. Simulation Settings

1) Data Generation: We collect $L = 24$ X-band data acquisitions at a center frequency of $f_c = 9.6$ GHz, which are distributed among a baseline of $\Delta b = 245$ m at a range of $r = 621$ km, corresponding to an elevation resolution of $\rho = 40$ m.

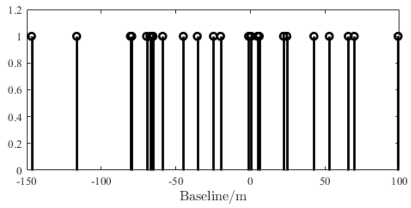

Fig. 3. Baseline distribution of the simulations.

The baseline distribution is shown in Fig. 3, and Fig. 4 presents the elevation beam pattern corresponding to this nonuniform baseline configuration. Due to this distribution, the ambiguity is suppressed. It can be observed that within the range of ±1000 m, no grating lobes with amplitudes comparable to the main lobe appear. Near 860 m, a highest sidelobe emerges, corresponding to the same position, where ambiguity occurs under uniform baseline distribution. This sidelobe has an amplitude of −7.2 dB. As a tradeoff, anomalous sidelobes occur. One can discover that sidelobes with −10 dB or so are located within $\pm(400-600)$ m. Also, the elevation resolution deteriorates. The 3-dB beamwidth is 41.3 m, corresponding to a resolution of 46.6 m, which represents a 16.5% loss compared to the theoretical resolution.

We set $N_t = 2$ point targets spaced at elevations of 10 and 50 m, respectively, with a distance equal to theoretical elevation resolution. The true positions of the targets are denoted by $s_{\text{true}} = 10, 50$ m. The SNR ranges from 0 to 40 dB. For each case, we execute $N_s = 5000$ Monte Carlo trials.

2) Evaluation Measurements: The estimation accuracy is characterized by two criteria: error and root mean square error (RMSE). Specifically, the error of one trial is defined as

$$\text{Error} = |\hat{s}_{q,\text{peak}} - s_{q,\text{true}}| \tag{35}$$

where $\hat{s}_{q,\text{peak}}$ and $s_{q,\text{true}}$ denote the estimates and the true values of the elevations of the qth target for $q = 1, 2, \ldots, N_t$, respectively. The RMSE is computed by

$$\text{RMSE} = \sqrt{\frac{\sum_{p=1}^{N_s}\sum_{q=1}^{N_t}\left(\hat{s}_{pq,\text{peak}} - s_{pq,\text{true}}\right)^2}{N_s N_t}} \tag{36}$$

where $\hat{s}_{pq,\text{peak}}$ and $s_{pq,\text{true}}$ denote the estimates and the true values of the elevations of the qth target in the pth trial, respectively.

B. Simulation Results

To present the performance of MLOS, we first give an intuitive comparison of MLOS and MARIA across all the trials. Then, we show the statistical results of MLOS and MARIA characterized by error and RMSE. At last, we show the statistical results of MLOS and CS characterized by RMSE.

1) Intuitive Comparison Versus MARIA: To evaluate the effect of two components of the MLOS method when the targets are closely spaced, we compare the following four methods, denoted by MARIA, MARIA with revised signal estimation (MARIA-RSE), as described in Section III-B, MARIA with iterative noise estimation (MARIA-INE) as Section III-C, and MLOS. The differences between these four methods are summarized in Table I. The results of all the Monte Carlo trials when SNR = 10, 20 dB are presented in Fig. 5, where the red solid lines represent the true positions of the targets.

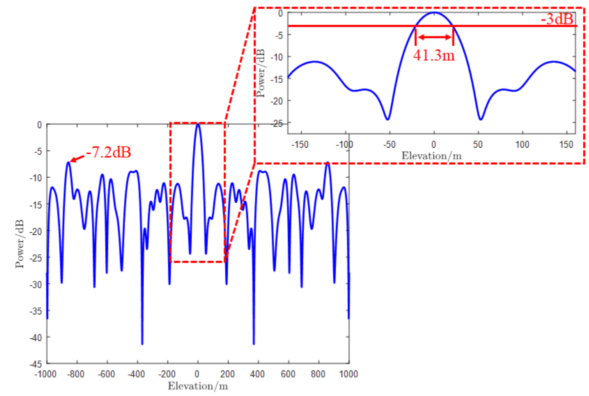

Fig. 4. Elevation beam pattern of the simulations.

TABLE I
METHODS FOR COMPARISON IN SIMULATION

Method	Signal estimation	Noise estimation
MARIA	error propagation	pre-defined
MARIA-INE	error propagation	iteratively estimate
MARIA-RSE	no error propagation	pre-defined
MLOS	no error propagation	iteratively estimate

We first take a look at the first row of Fig. 5, corresponding to the case when SNR = 20 dB. From Fig. 5(a) and (b), it can be observed that the results from MARIA and MARIA-INE show fluctuations distributed on both sides of the red lines. The two methods show little difference across all the trials, suggesting that whether noise is iteratively estimated or not has negligible impact on performance. This can be attributed to the fact that the iteration in MARIA is driven by positive feedback, where noise estimation plays a minor role. In Figs. 5(c) and 5(d), the targets are closer to the two red lines than in Fig. 5(a) and (b), indicating that the RSE method demonstrates greater robustness to noise compared to MARIA. This further verifies that the removal of positive feedback improves accuracy. By comparing Figs. 5(d) and 5(c), it can be observed that the targets tend to be distributed along the inner sides of the two red lines by MARIA-RSE, whereas this tendency is less pronounced in MLOS. This reflects the significance of noise estimation. When targets are correctly identified within a reasonable region, an accurate noise estimation enhances the precision of signal estimation. From the results above, the accuracy of the RSE method is validated through this high SNR experiment, while the effect of INE needs to be further assessed in a lower SNR experiment.

Next, we analyze the second row of Fig. 5, corresponding to the case when SNR = 10 dB. From Fig. 5(e) and (f), it is evident that the iterations are significantly influenced for MARIA and MARIA-INE by noise. Most results deviate from the red solid lines, with more pronounced vertical spreading compared to the 20-dB case. This is because noise further disrupts signal estimation during iterations, leading to an increased number of outliers. In Fig. 5(g) and (h), targets are

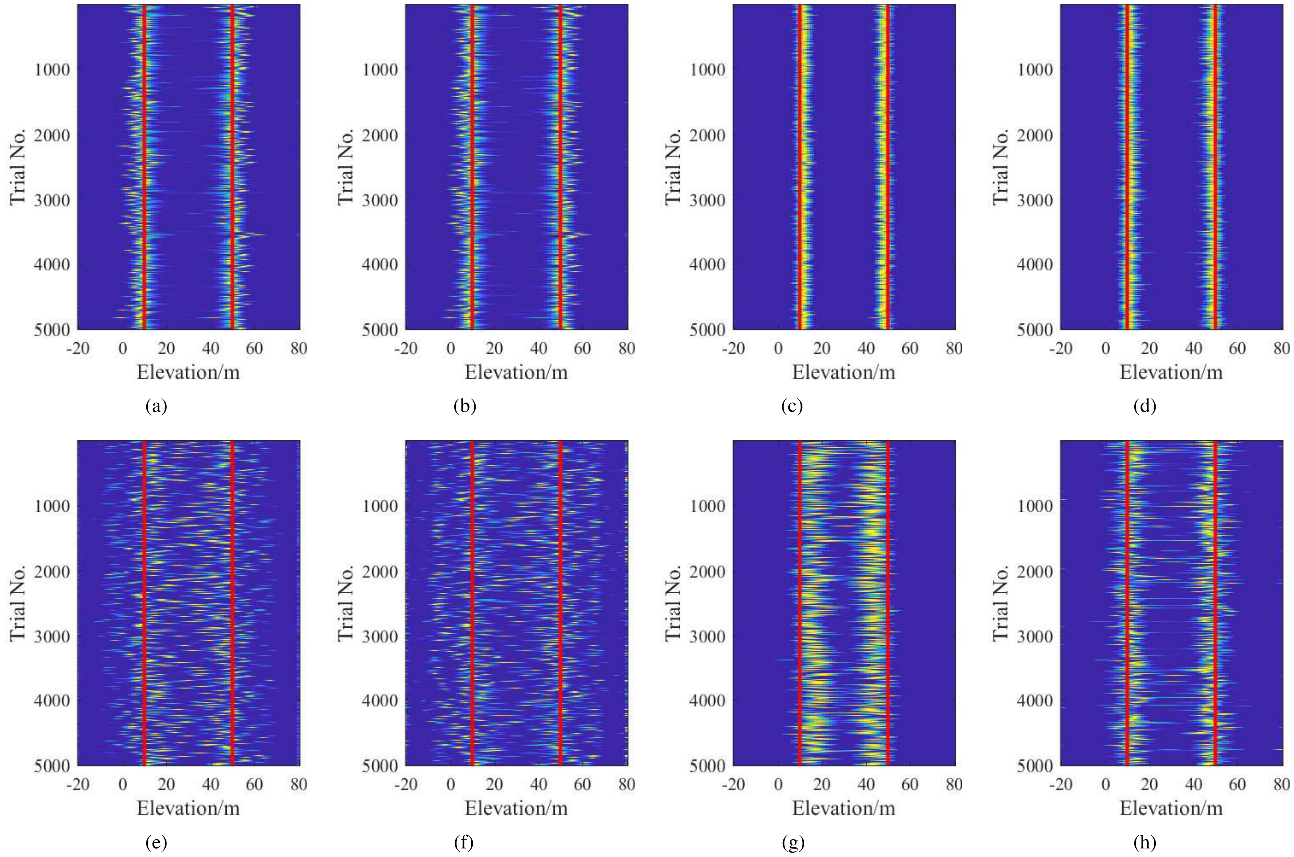

Fig. 5. Simulation results of double-point targets by different methods at different SNR levels. Red solid lines are the true positions. From (left to right): MARIA, MARIA-INE, MARIA-RSE, and MLOS. (a) Method: MARIA, SNR = 20 dB. (b) Method: MARIA-INE, SNR = 20 dB. (c) Method: MARIA-RSE, SNR = 20 dB. (d) Method: MLOS, SNR = 20 dB. (e) Method: MARIA, SNR = 10 dB. (f) Method: MARIA-INE, SNR = 10 dB. (g) Method: MARIA-RSE, SNR = 10 dB. (h) Method: MLOS, SNR = 10 dB.

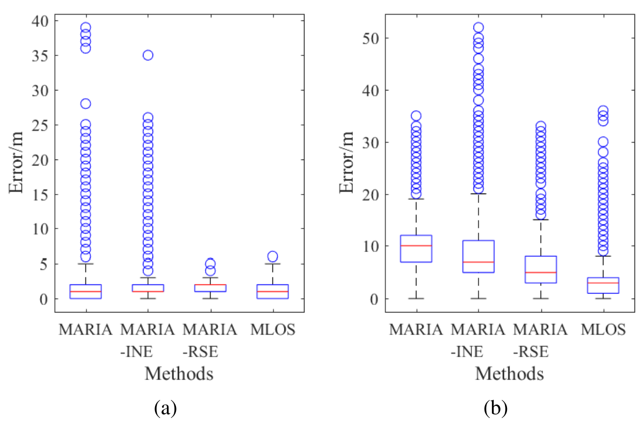

Fig. 6. Boxplots of the simulation results at different SNR levels. Methods in each boxplot from (left to right): MARIA, MARIA-INE, MARIA-RSE, and MLOS. The red lines in the blue boxes are medians of the results, and the blue boxes indicate the interquartile range, spanning from the lower quartile to the upper quartile. The blue circles are outliers, which are identified when the error is 1.5× interquartile range higher than the upper quartile. Two short horizontal lines are the nonoutlier maximum and minimum. (a) SNR = 20 dB. (b) SNR = 10 dB.

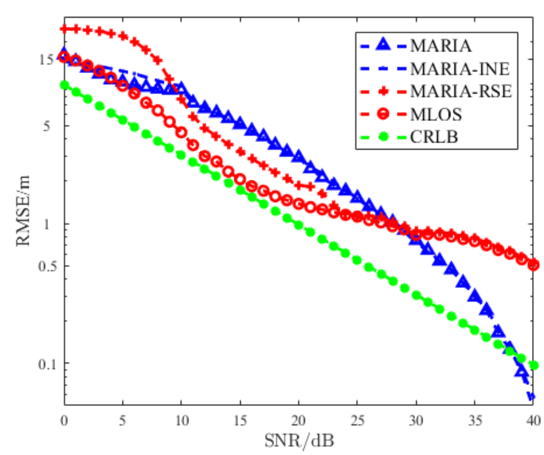

Fig. 7. RMSEs of the four methods and Cramér-Rao lower bound (CRLB) with respect to SNR.

noticeably closer to the red lines than the other two methods, which show the robustness of the RSE method under low SNR conditions. By comparing Fig. 5(g) and (h), a similar phenomenon occurs more obviously than when SNR = 20 dB, with almost all targets falling within the interval between the two red lines by MARIA-RSE, which indicates targets are not fully separated. In contrast, results by MLOS exhibit better separation. This reveals the INE method enhances signal estimation accuracy. Although outliers are present, the MLOS method exhibits the best performance under low SNR conditions.

2) Statistical Results Versus MARIA: We statistically analyze the algorithm performance from the aspect of error and RMSE. The boxplots of errors for the four methods when SNR = $10, 20$ dB are shown in Fig. 6. When SNR = 20 dB, all the four methods yield errors below 5 m in most trials. However, MARIA and MARIA-INE show significantly

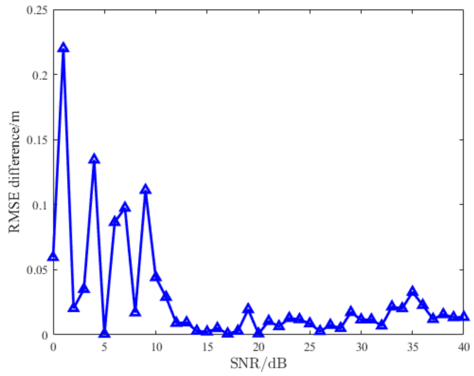

Fig. 8. RMSE difference between beamforming and uniform distribution vector with respect to SNR.

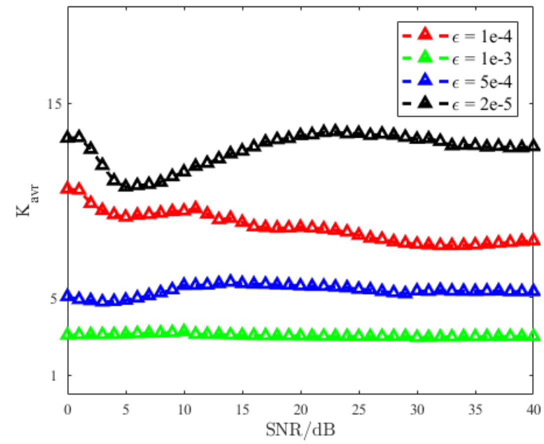

Fig. 9. Average number of iterations K_{avr} of different ϵ selections with respect to SNR.

more outliers and larger errors compared to MARIA-RSE and MLOS. This verifies the effect of the RSE method. Between methods with the same signal estimation method, MLOS has a lower median and lower quartile than MARIA-RSE. This indicates that MLOS produces more targets precisely on the red lines than MARIA-RSE, where the elevations of the targets are correctly estimated. When SNR = 10 dB, MARIA exhibits the highest median. MARIA-INE has a smaller median and interquartile range compared to MARIA, but it also produces the largest outlier. MARIA-RSE outperforms the former two methods across all indices, while MLOS surpasses MARIA-RSE in all metrics except for outliers. The effect of INE is particularly evident in this boxplot. While noise estimation exacerbates inaccuracies when targets are significantly deviated, it improves the accuracy when the signal estimation falls within a reasonable range, resulting in an overall positive effect.

The outliers in the boxplots highlight the discrepancy between the theoretical convergence analysis and practical results. The theoretical convergence is derived under the assumptions of a column-orthogonal measurement matrix and uniform noise power. However, these assumptions are rarely strictly satisfied in practice, which can lead to incomplete convergence of the iterative process for the following reasons: 1) the column orthogonality assumption does not strictly hold, as our derivation only shows that the correlation between two columns of the matrix $\mathbf{A}$ is probabilistically bounded rather than exactly zero. In particular, under nonuniform baseline configurations, adjacent baselines tend to produce higher column correlations, which may impair the convergence performance and 2) the assumption of identical noise power is also violated in practice. This discrepancy can be amplified during the matrix inversion process, thereby affecting the convergence behavior and estimation accuracy.

To further compare the accuracy of the four methods statistically, the RMSEs at different SNR levels for different methods are illustrated in Fig. 7. When SNR > 5 dB, MLOS achieves similar performance with MARIA, while MARIA-RSE performs the worst. As the SNR increases from 5 to 10 dB, both MLOS and MARIA-RSE exhibit a rapid decline in RMSE, placing them among the best. In contrast, MARIA and MARIA-INE experience only a modest 3-m decrease in RMSE within the same range. When SNR > 10 dB, the RMSEs for all the four methods tend to follow the same decreasing trend as the CRLB in logarithmic coordinates. Especially, there are obvious gaps between MARIA, MARIA-RSE, and MLOS. Meanwhile, MARIA and MARIA-INE show almost identical performance. The difference of MARIA-RSE and MLOS is due to the overall positive effect of INE. The identical performance of MARIA and MARIA-INE can be explained by the minor impact of noise estimation on MARIA. As the SNR exceeds 20 dB, the performance gap among the four methods diminishes, with all methods exhibiting comparable results at an SNR of 30 dB. When SNR > 30 dB, MARIA and MARIA-INE exhibits a rapid decrease in RMSE, even better than CRLB. This demonstrates that the positive feedback mechanism of MARIA is well suited for high SNR scenarios. The facts above show that the RSE method consistently outperforms MARIA when the SNR ranges from 5 to 20 dB. Furthermore, the comparison between MARIA-RSE and MLOS highlights the significance of the iterative noise power estimation. The results also suggest that neither MLOS nor MARIA is effective when the SNR falls below 5 dB. For a stack of SAR images within this SNR range, the targets are no longer distinguishable from the surrounding noise, making it extremely challenging to perform 3-D imaging on such low-quality SAR data. The failure of both methods in this SNR range is therefore sensible and expected. When the SNR exceeds 20 dB, the positive feedback mechanism in MARIA becomes more effective, leading to improved estimation accuracy. In contrast, the modifications introduced by MLOS provide no significant improvement in high SNR scenarios. When SNR > 30 dB, MARIA outperforms MLOS, even exceeding the CRLB, likely due to the effective utilization of positive feedback in high SNR cases. It should be noted that 30 dB is actually a high SNR level, which lies beyond the scope of low SNR in this study.

3) Hyperparameter Sensitivity: To evaluate the iterative robustness of the MLOS method under different hyperparameter settings, this section analyzes its sensitivity to initial conditions, number of iterations, and convergence threshold. Under ideal conditions, according to the fixed-point theorem and Proposition 1, poor initial values increase the number of iterations required for convergence but do not affect the final

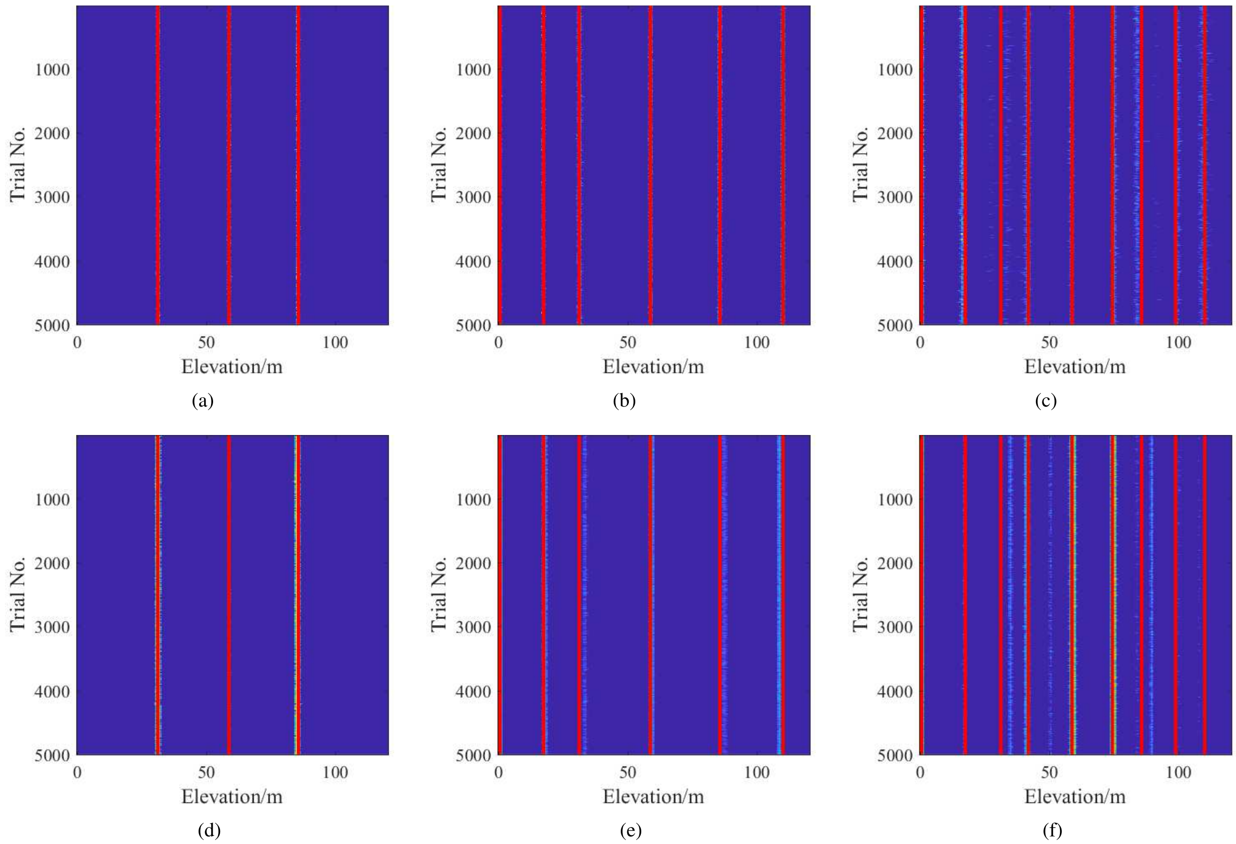

Fig. 10. Upper row are the results by MLOS and the lower row are by ISTA. Here, (a) and (d) are N_t = 3, (b) and (e) are N_t = 6, and (c) and (f) are N_t = 9.

TABLE II
POSITIONS OF TARGETS WHEN N_t INCREASES FROM 1 TO 10

Targe No.	1	2	3	4	5
Elevation/m	31	86	59	1	110
Targe No.	6	7	8	9	10
Elevation/m	18	74	42	99	115

convergence result. In practice, the ideal assumptions may not be strictly satisfied, which reduces the algorithm's robustness during iteration.

In order to analyze the impact of different initialization strategies on the iterative process, two sets of initial conditions are applied to the data. The first uses the beamforming output as the initial estimate, while the second initializes with a random vector generated by a uniform distribution. The RMSEs under both initialization strategies are compared. Fig. 8 illustrates the RMSE difference between the two initialization schemes across the SNR range. It can be observed that within the 0–40-dB SNR range, the maximum RMSE difference is 0.22 m, with an average difference of 0.03 m. This indicates that the proposed method remains consistent with beamforming initialization, even under poor initial conditions.

In order to analyze the convergence tolerance sensitivity, we focus on analyzing the average number of iterations K_{avr} required for convergence under different values of ϵ. Fig. 9 presents the convergence curves of the method under different values of ϵ. It can be observed that across various SNR levels, MLOS achieves convergence when $\epsilon = 5 \times 10^{-4}$ with about five iterations. When $\epsilon = 1 \times 10^{-4}$, the number of iterations required for convergence generally decreases with increasing SNR. Once SNR exceeds 10 dB, the average number of iterations falls between 6 and 8, indicating that $\epsilon = 1 \times 10^{-4}$ is an appropriate choice. This also suggests that higher SNR facilitates faster convergence. In contrast, when $\epsilon = 2 \times 10^{-5}$, the iteration count approaches $K_{\max}$, indicating that the threshold is overly strict and that such high precision significantly increases computational cost.

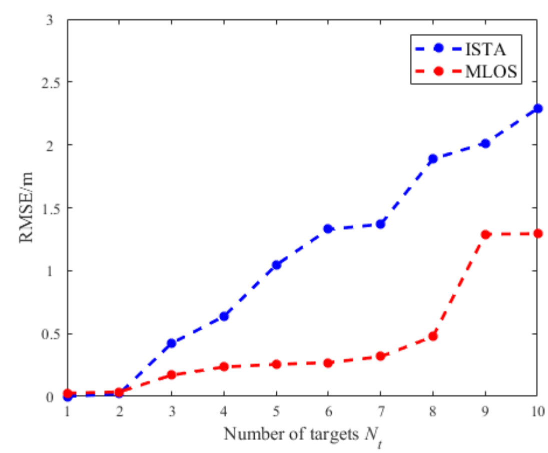

Fig. 11. RMSEs of ISTA and MLOS with respect to N_t.

4) Statistical Results Versus CS: In order to evaluate the performance of MLOS from sparse to nonsparse target distributions, we compare it with iterative shrinkage and thresholding algorithm (ISTA) by varying the number of

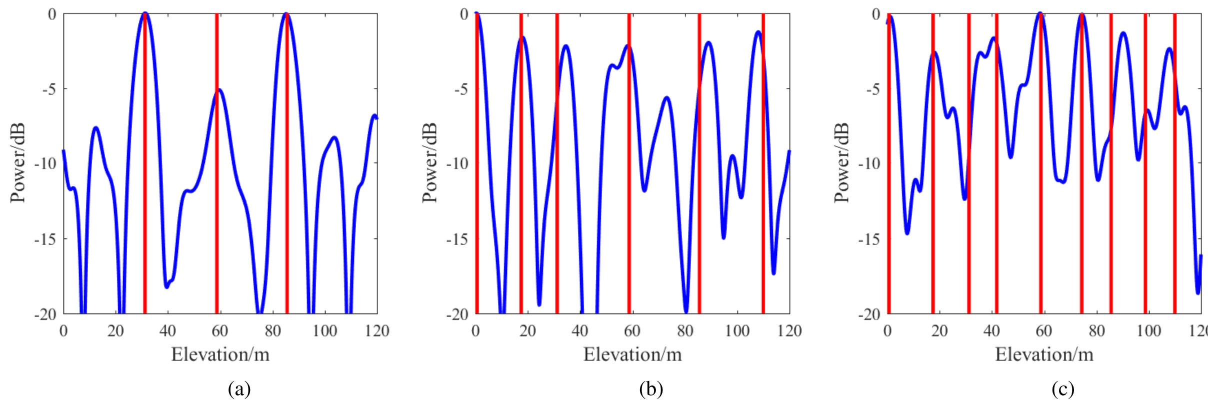

Fig. 12. Response of the beam pattern when (a)–(c) $N_t = 3, 6,$ and 9.

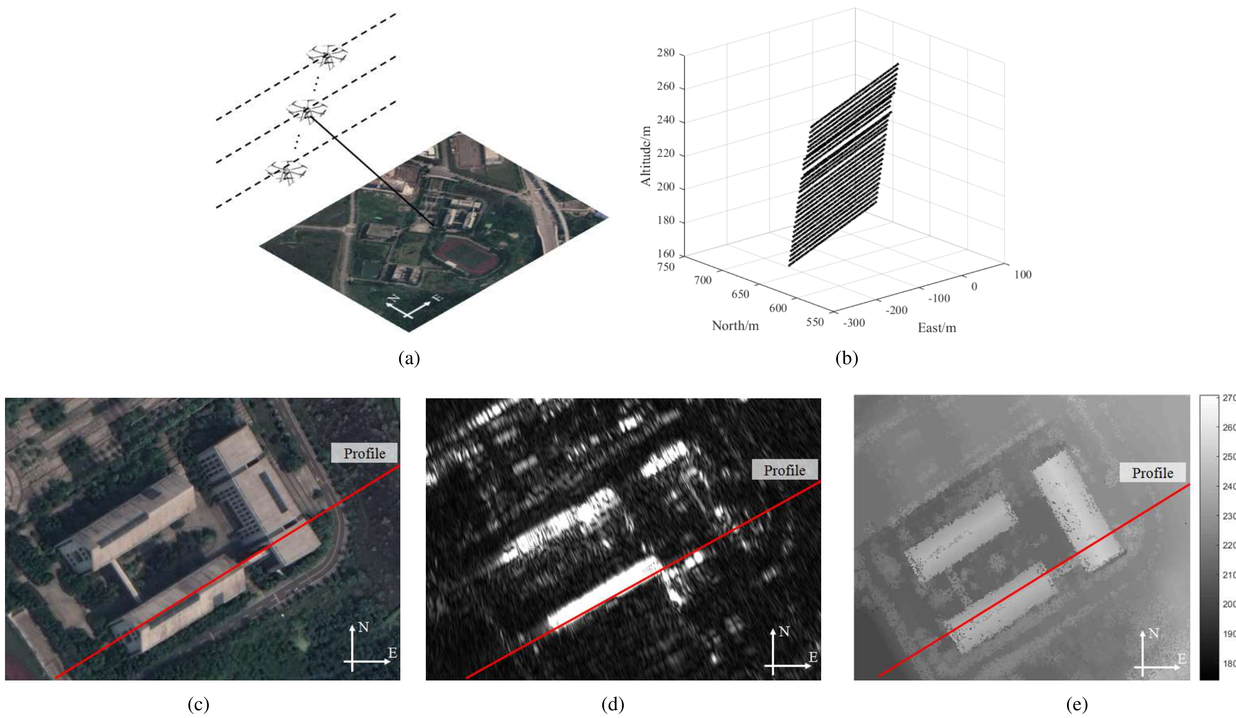

Fig. 13. Real UAV TomoSAR 3-D experiment settings. (a) Data acquisition geometry. (b) Baseline configuration. (c) Optical image. (d) 2-D radar image. (e) DSM.

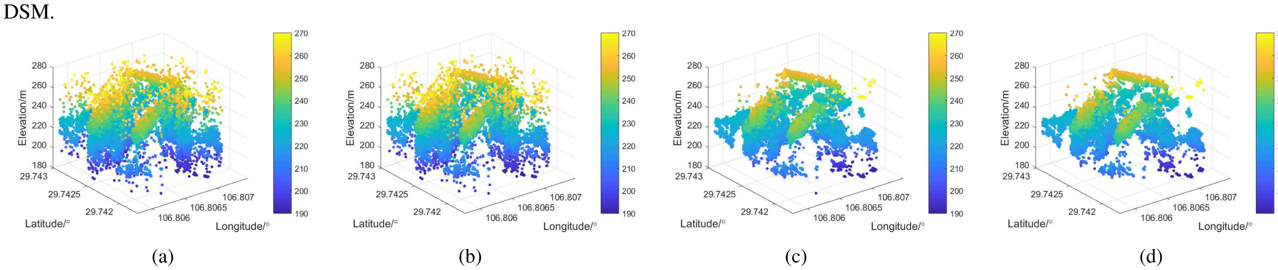

Fig. 14. TomoSAR 3-D pointcloud images of the region of interest by different methods on the original data. (a) MARIA. (b) MARIA-INE. (c) MARIA-RSE. (d) MLOS.

targets. Parameters are identical to the former one except for the overall baseline, elevation resolution, target distribution, and SNR. The overall baseline increases to 1960 m, yielding elevation resolution to 5 m. Targets are randomly distributed in the whole unambiguous range $H_{\text{amb}} = 120$ m, as computed in (6). The number of targets ranges from 1 to 10. Table II gives the positions of the targets, with the interval between any two targets exceeding resolution. When N_t increases to k, the kth target is incorporated. The SNR is fixed to be 20 dB. Fig. 10 presents the results of all Monte Carlo trials for $N_t = 3$, 6, and 9, where the red lines indicate the true positions of the targets. It can be observed that when $N_t = 3$, both MLOS and ISTA are able to reasonably recover the target locations, although ISTA begins to show noticeable deviation in estimating the rightmost target. When $N_t = 6$, MLOS continues to provide accurate estimates, while ISTA's results deviate significantly

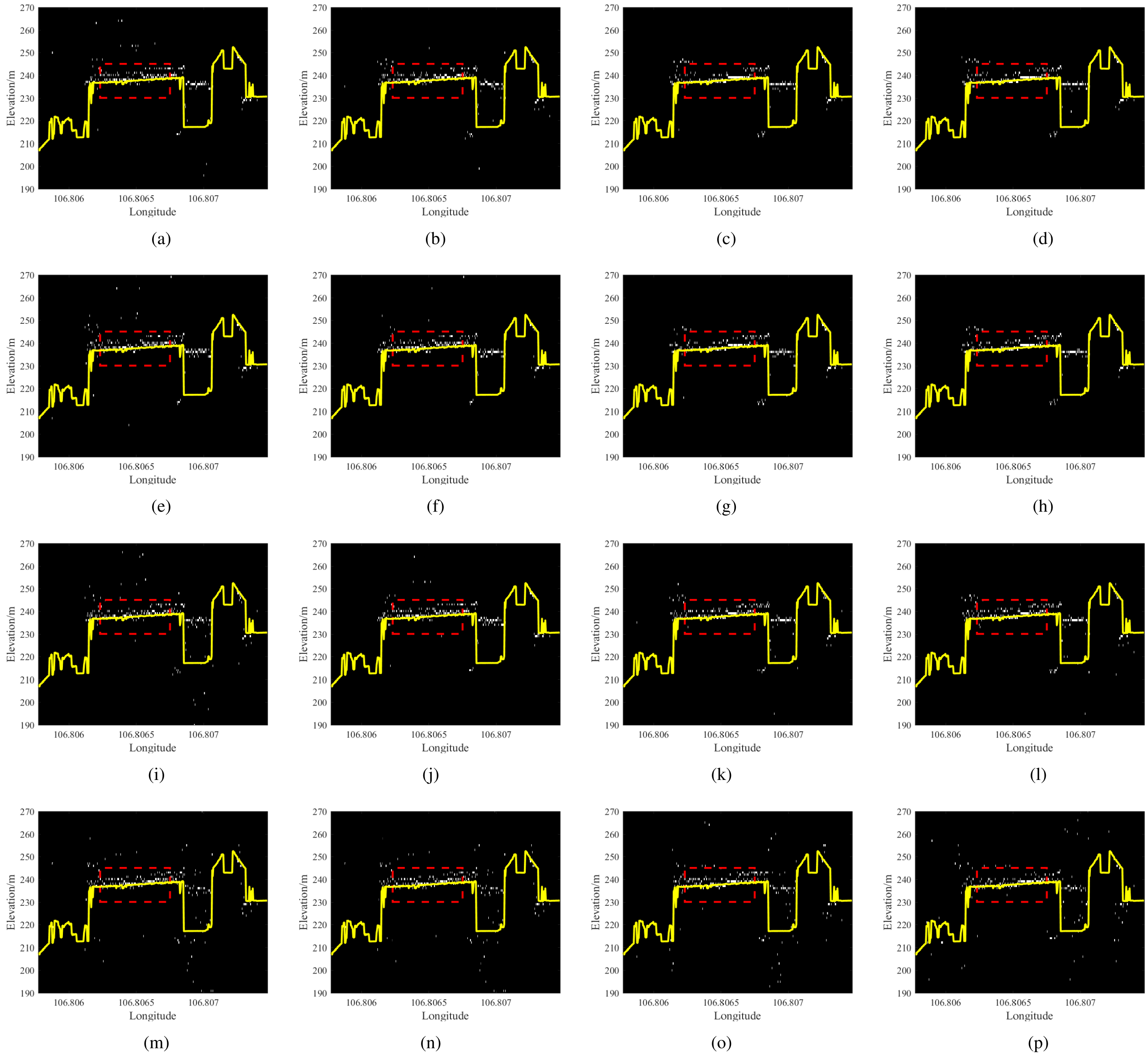

Fig. 15. TomoSAR 3-D images of the profile achieved by different methods at different $\mathrm{SNR}_{\mathrm{loss}}$ levels. (a) MARIA, $\mathrm{SNR}_{\mathrm{loss}} = 0$ dB. (b) MARIA-INE, $\mathrm{SNR}_{\mathrm{loss}} = 0$ dB. (c) MARIA-RSE, $\mathrm{SNR}_{\mathrm{loss}} = 0$ dB. (d) MLOS, $\mathrm{SNR}_{\mathrm{loss}} = 0$ dB. (e) MARIA, $\mathrm{SNR}_{\mathrm{loss}} = 5$ dB. (f) MARIA-INE, $\mathrm{SNR}_{\mathrm{loss}} = 5$ dB. (g) MARIA-RSE, $\mathrm{SNR}_{\mathrm{loss}} = 5$ dB. (h) MLOS, $\mathrm{SNR}_{\mathrm{loss}} = 5$ dB. (i) MARIA, $\mathrm{SNR}_{\mathrm{loss}} = 10$ dB. (j) MARIA-INE, $\mathrm{SNR}_{\mathrm{loss}} = 10$ dB. (k) MARIA-RSE, $\mathrm{SNR}_{\mathrm{loss}} = 10$ dB. (l) MLOS, $\mathrm{SNR}_{\mathrm{loss}} = 10$ dB. (m) MARIA, $\mathrm{SNR}_{\mathrm{loss}} = 15$ dB. (n) MARIA-INE, $\mathrm{SNR}_{\mathrm{loss}} = 15$ dB. (o) MARIA-RSE, $\mathrm{SNR}_{\mathrm{loss}} = 15$ dB. (p) MLOS, $\mathrm{SNR}_{\mathrm{loss}} = 15$ dB.

from the true positions. At $N_t = 9$, both methods yield evidently incorrect estimates, leading to a substantial increase in RMSE. Seen from Fig. 11, when the number of targets is no more than two, ISTA can give an outstanding result, outperforming MLOS. However, as the number of targets increases, MLOS exhibits greater robustness, achieving lower RMSE compared to ISTA. The sudden increase when N_t increases from 2 to 3 and from 7 to 8 reveals ISTA fails to distinguish targets accurately, while MLOS does not exhibit a similar phenomenon until N_t increases from 8 to 9. The sudden increase is attributed to the complicated response of the nonuniform baseline distribution. Fig. 12 shows the response of the beam pattern when $N_t = 3, 6$, and 9. As N_t increases, response extrema tend to deviate from the true target positions. Especially, when $N_t = 9$, the response at 31 m is the lowest among all nine targets and an extremum at 25 m also misleads the target position estimation. Both factors make it hard to recover the target correctly. These results indicate that MLOS is comparable to ISTA in sparse cases and is more capable for a nonsparse scenario than CS methods.

C. UAV TomoSAR Experiment

1) Experimental Settings: To test the performance of the proposed MLOS method in real data, a UAV TomoSAR experiment is executed, in which we focus on a building area. The TomoSAR 3-D imaging experiment [33] using real UAV SAR data was conducted at Chongqing Aeronautical Institute, China. The UAV platform operates in P-band, achieving a resolution of 1 m in both range and azimuth. The experiment includes 30 baselines, with heights ranging from 170 to 260 m, yielding a 3-m elevation resolution. Fig. 13(a) and (b)

Fig. 16. TomoSAR 3-D pointclolud image of the entire illuminated area.

TABLE III
COMPARISON OF THE RMSES IN FIG. 15

SNR_{loss}	MARIA	MARIA-INE	MARIA-RSE	MLOS
0dB	5.34m	2.41m	2.10m	1.96m
5dB	5.38m	3.42m	2.23m	2.11m
10dB	5.49m	4.28m	2.43m	2.20m
15dB	7.88m	4.75m	4.41m	2.89m

illustrates the observation geometry and baseline configuration, respectively. To evaluate the performances under varying noise levels using real data, the dataset is divided into four groups. The first group uses the original data, while the other three groups are corrupted with additional noise power 5, 10, and 15 dB higher than the original data. When adding noise, the average noise power of the 2-D SAR image can be estimated from the amplitude histogram of the image. The amplitude of a complex Gaussian noise with variance σ^2 for both real and imaginary parts satisfies Rayleigh distribution, whose maximum probability density appears at σ. Therefore, we plot the amplitude histogram of the SAR image and identify the peak location, denoted as R_{peak}, which can be interpreted as the estimate of σ and the average noise power can be estimated as $2R_{\text{peak}}^2$. Then, according to the estimated noise level, the complex Gaussian noise can be added proportionally. These four groups are referred to as $\text{SNR}_{\text{loss}} = 0, 5, 10$, and 15 dB in the subsequent analysis, respectively. The region of interest encompasses three buildings. Two of them are aligned in an east–west direction, while the other one is positioned perpendicular to them. The optical image, 2-D radar image, and the digital surface model (DSM) achieved by LiDAR are showed as Fig. 13(c)–(e), respectively. The profile for analysis is marked by a red straight line, crossing the south building in east–west direction. Tracks are located to the north of the target scene, moving from west to east, with the antenna beam directing toward the south.

2) Results: Fig. 14 shows the 3-D pointcloud images of the region of interest by MARIA, MARIA-INE, MARIA-RSE, and MLOS on the original data. A notable distinction among the four methods is that the MARIA and MARIA-INE produce significantly more outliers compared to MARIA-RSE and MLOS. The reconstructed elevations of these outliers deviate from their true positions by over 10 even 20 m. This verifies the results in simulations that, for high SNR cases, the application of the RSE method is much more effective than MARIA in suppressing outliers in real-world scenarios.

For quantitive analysis, we give the results of the selected profile by MARIA, MARIA-INE, MARIA-RSE, and MLOS, in Fig. 15, in which the yellow solid lines represent the ground truth given by DSM data. All the four methods perform worse as SNR decreases. For the original data, MARIA produces the most outliers compared to the other three methods. While MARIA-INE can suppress most outliers, it struggles to recover a flat eave. The results by MARIA-RSE and MLOS are much flatter, which fit the actual shape of an eave. Most points from MARIA-RSE and MLOS are within 2 m above the ground truth, whereas MARIA and MARIA-INE contain several points deviating from the true elevation by 3 m or even more. When the data are corrupted by 5-dB noise, MARIA produces more outliers, and MARIA-INE is also less effective at suppressing them. MARIA-RSE and MLOS still manage to suppress most outliers, though the results are not as flat as those on the original data. When $\text{SNR}_{\text{loss}} = 10$ dB, maintaining a flat shape becomes more difficult for all methods. MARIA-RSE and MLOS still outperform MARIA-INE and MARIA. When $\text{SNR}_{\text{loss}} = 15$ dB, outliers are prominent among all the four methods. Along the selected eave, MARIA produces the most outliers, followed by MARIA-INE, with MLOS performing the best. Results are consistent with simulations that MARIA and MARIA-INE are not robust in signal estimation, leading to outliers, whereas MLOS effectively addresses this issue.

Table III presents the RMSE comparison computed from the section outlined by rectangles with red dashed border. MARIA gives the worst result at all four SNR levels. Compared with MARIA, MARIA-INE reduces more RMSE in real experiment than in simulations, primarily because the larger elevation span leads to larger errors. When $\text{SNR}_{\text{loss}} \leq 10$ dB, the gap between MLOS and MARIA-RSE is less than 11%, which becomes more than 50% when $\text{SNR}_{\text{loss}} = 15$ dB. The RMSE by MARIA-RSE is close to MARIA-INE when $\text{SNR}_{\text{loss}} = 15$ dB, which indicates both methods tend to fail at this SNR level, while MLOS is still robust to noise. This comparison verifies the outstanding performance of MLOS among the four methods. Fig. 16 shows the 3-D pointcloud image of the entire illuminated area by MLOS, superimposed on the optical image.

VI. Conclusion

In this article, in order to improve the TomoSAR 3-D imaging accuracy especially in low SNR scenarios, we proposed a novel TomoSAR 3-D imaging method, MLOS. Specifically, two main ideas were proposed to improve the applicability: first, an RSE method without positive feedback mechanism was developed, increasing the robustness in low SNR; second, an INE method was proposed to address cases where the pre-estimated noise power becomes inaccurate. Then, the effectiveness and advantages of MLOS were validated through simulations and real UAV experiments. Finally, a theoretical analysis was provided to demonstrate the convergence guarantee for the proposed MLOS method.

Appendix A
Proof of Proposition 1

We use the fixed-point theorem [34] to prove that the iteration of $\mathbf{v}$ in (23) converges.

First, we simplify the expression of $\mathbf{v}$ in (23) based on the assumptions. For $\mathbf{F}$ in (19), we have

$$\mathbf{F} = \mathbf{A}^{+}\mathbf{R}_{\mathbf{g}}^{-1} = \mathbf{A}^{+}\left(D(\mathbf{u}) + \mathbf{A}D(\mathbf{v})\mathbf{A}^{+}\right)^{-1}. \tag{A.1}$$

Through the following matrix inversion formula:

$$(\mathbf{A}+\mathbf{BCD})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{B}\left(\mathbf{C}^{-1}+\mathbf{DA}^{-1}\mathbf{B}\right)^{-1}\mathbf{DA}^{-1}. \tag{A.2}$$

We rewrite (A.1) into

$$\mathbf{F} = \left(\mathbf{I} - \mathbf{W}\left(D(\mathbf{v})+\mathbf{W}\right)^{-1}\right)\mathbf{A}^{+}D^{-1}(\mathbf{u}) \tag{A.3}$$

where $\mathbf{W} = \mathbf{A}^{+}D^{-1}(\mathbf{u})\mathbf{A}$.

Based on the assumptions that $\mathbf{A}^{+}\mathbf{A} = L\mathbf{I}$ and $D(\mathbf{u}) = \sigma^2\mathbf{I}$, we have $\mathbf{W} = \sigma^{-2}L\mathbf{I}$. Then, (A.3) is rewritten as

$$\mathbf{F} = D(\mathbf{c})\mathbf{A}^{+} \tag{A.4}$$

where $[\mathbf{c}]_m = ([\mathbf{v}]_m/\sigma^2[\mathbf{v}]_m + L), m = 1,\ldots,M$. Substituting (A.4) to (23) implies that

$$\mathbf{v} = \mathbf{P}^{-1}\left\{D(\mathbf{c})\mathbf{A}^{+}\left(\mathbf{gg}^{+}-\sigma^2\mathbf{I}\right)\mathbf{A}D(\mathbf{c})\right\}_{\text{diag}} \tag{A.5}$$

where $\mathbf{P} = L^2D^2(\mathbf{c})$. Since $D(\mathbf{c})$ is a diagonal matrix, (A.5) can be written as

$$\begin{aligned}\mathbf{v} &= \mathbf{P}^{-1}D^2(\mathbf{c})\left\{\mathbf{A}^{+}\left(\mathbf{gg}^{+}-\sigma^2\mathbf{I}\right)\mathbf{A}\right\}_{\text{diag}}\\ &= \frac{1}{L^2}\left\{\mathbf{A}^{+}\left(\mathbf{gg}^{+}-\sigma^2\mathbf{I}\right)\mathbf{A}\right\}_{\text{diag}}.\end{aligned} \tag{A.6}$$

Note that σ^2 in (A.6) actually corresponds to the eigenvalues of the residual matrix. Since the sum of the eigenvalues of a matrix equals the trace of the matrix, σ^2 is estimated by

$$\sigma^2 = \frac{1}{L}\text{trace}\left(\mathbf{gg}^{+} - \mathbf{A}D(\mathbf{v})\mathbf{A}^{+}\right). \tag{A.7}$$

Substituting (A.7) to (A.6) implies

$$\mathbf{v} = \boldsymbol{\varphi}(\mathbf{v}) \tag{A.8}$$

where

$$\boldsymbol{\varphi}(\mathbf{v}) = \frac{1}{L^3}\left\{L\mathbf{A}^{+}\mathbf{gg}^{+}\mathbf{A} - \mathbf{A}^{+}\text{trace}\left(\mathbf{gg}^{+}-\mathbf{A}D(\mathbf{v})\mathbf{A}^{+}\right)\mathbf{A}\right\}_{\text{diag}}. \tag{A.9}$$

The fixed-point theorem states that

Theorem 1: For $\varphi(x) \in C[x_1, x_2]$, if 1) $x \in [x_1, x_2], x_1 \leqslant \varphi(x) \leqslant x_2$ and 2) $\exists\eta > 0, 0 < \eta < 1, |\varphi'(x)| < \eta$, we have the following.

1) $x = \varphi(x)$ has a unique solution $x^* \in [x_1, x_2]$.
2) The iteration $x_{k+1} = \varphi(x_k)$ converges to x^*.

We apply the fixed-point theorem to prove the convergence of the iteration in (A.8). Since $\mathbf{v}, \boldsymbol{\varphi}(\mathbf{v}) > \mathbf{0}$ and $\mathbf{v}, \boldsymbol{\varphi}(\mathbf{v})$ are all bounded, we let $x_1 = 0, x_2 = \max\{\mathbf{v}, \boldsymbol{\varphi}(\mathbf{v})\}$, where the second term indicates the maximum element of the vectors. For any $[\mathbf{v}]_m, m = 1,\ldots,M$, we have

$$\begin{aligned}\frac{\partial\left[\boldsymbol{\varphi}(\mathbf{v})\right]_m}{\partial[\mathbf{v}]_m} &= \frac{1}{L^3}\frac{\partial\left[\left\{\mathbf{A}^{+}\text{trace}\left(\mathbf{A}D(\mathbf{v})\mathbf{A}^{+}\right)\mathbf{A}\right\}_{\text{diag}}\right]_m}{\partial[\mathbf{v}]_m}\\ &= \frac{1}{L^2}\frac{\partial\ \text{trace}\left(\mathbf{A}D(\mathbf{v})\mathbf{A}^{+}\right)}{\partial[\mathbf{v}]_m}\\ &= \frac{1}{L}\frac{\partial\ \text{trace}\left(D(\mathbf{v})\right)}{\partial[\mathbf{v}]_m}\\ &= \frac{1}{L} < 1.\end{aligned} \tag{A.10}$$

Based on Theorem 1, (A.10) means that the iteration in (A.8) converges, yielding the complete of the proof.

Appendix B
Proof of Proposition 2

The proof is based on the Hoeffding inequality [35]. Particularly, we first rewrite $[\mathbf{A}^{+}\mathbf{A}]_{m_1,m_2}$ into

$$\begin{aligned}\frac{1}{L}\left[\mathbf{A}^{+}\mathbf{A}\right]_{m_1,m_2} &= \frac{1}{L}\sum_{l=1}^{L}\exp\left(j\Delta\omega b_l\right)\\ &= \frac{1}{L}\left(\sum_{l=1}^{L}\cos\Delta\omega b_l + j\sum_{l=1}^{L}\sin\Delta\omega b_l\right).\end{aligned} \tag{B.1}$$

When $b_l \in \mathcal{U}[0, D]$, we have

$$\mathbb{E}\left(\frac{1}{L}\sum_{n=1}^{L}\cos\Delta\omega b_l\right) = \frac{\sin\Delta\omega D}{\Delta\omega D} \tag{B.2}$$

$$\mathbb{E}\left(\frac{1}{L}\sum_{n=1}^{L}\sin\Delta\omega b_l\right)=\frac{1-\cos\Delta\omega D}{\Delta\omega D}. \tag{B.3}$$

Through the Hoeffding inequality and some straightforward derivation, we have

$$p_1 \equiv p\left(\left|\frac{1}{L}\sum_{l=1}^{L}\cos\Delta\omega b_l\right| \geqslant t_1\right) \leqslant 2e^{-\frac{Lt^2}{2}} \tag{B.4}$$

$$p_2 \equiv p\left(\left|\frac{1}{L}\sum_{l=1}^{L}\sin\Delta\omega b_l\right| \geqslant t_2\right) \leqslant 2e^{-\frac{Lt^2}{2}} \tag{B.5}$$

where t_1, t_2 are defined in (32) and (33), respectively. For $[\mathbf{A}^{+}\mathbf{A}]_{m_1,m_2}$, we have

$$p\left(|\frac{1}{L}\left[\mathbf{A}^{+}\mathbf{A}\right]_{m_1,m_2}| \geqslant \sqrt{t_1^2+t_2^2}\right) \leqslant p_1+p_2 \leqslant 4e^{-\frac{Lt^2}{2}}. \tag{B.6}$$

Completing the proof.

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